Data On Swiss Residential PV Generation

Pitfalls of joining several open data sources

Raul Hochul, University of Basel, raul.hochuli@unibas.ch

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solkat solkat month gwr pv

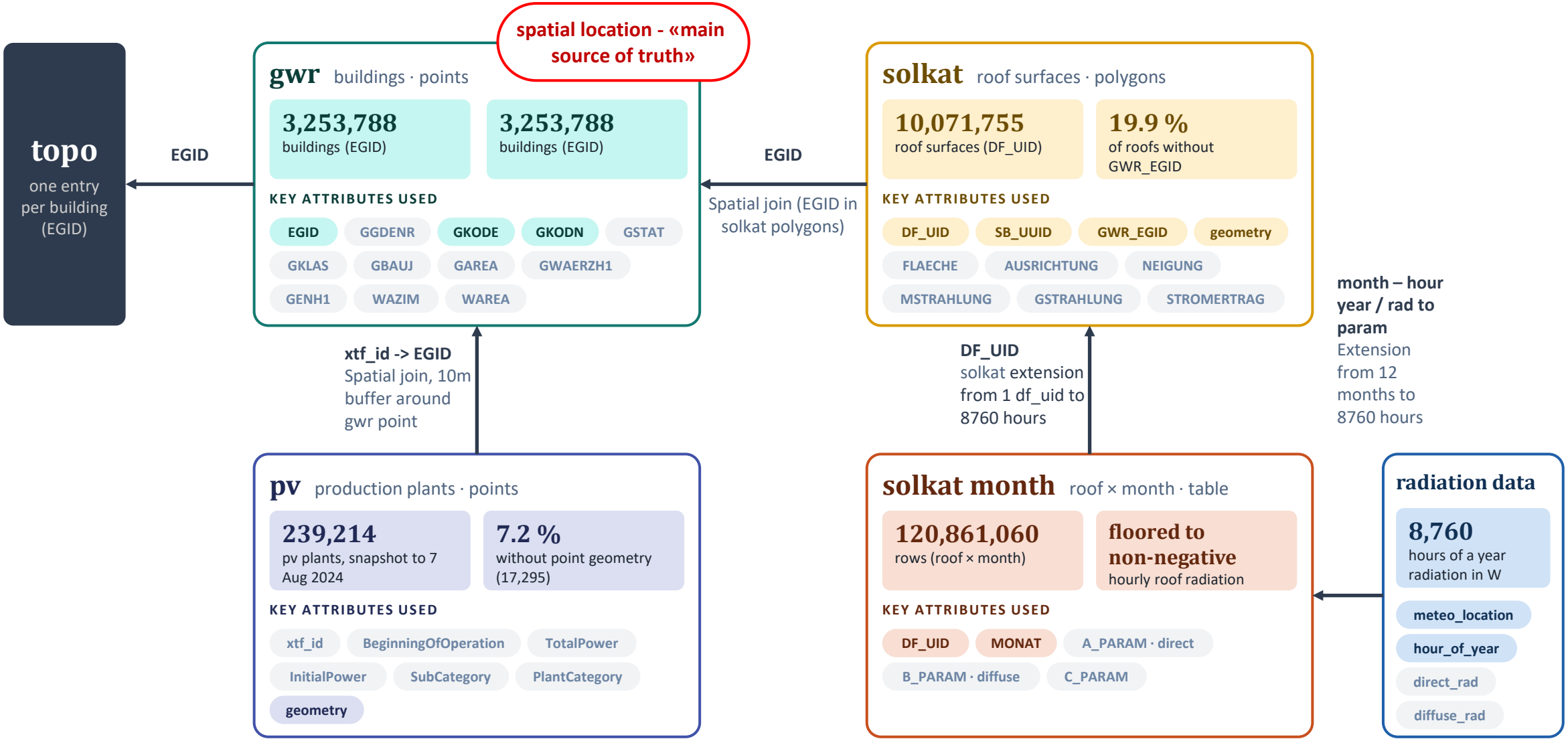
01

Data Sources

Four sources, four units of observation

Source	Unit of observation	Rows, CH-wide	Data Source	Size on local disk
solkat **(solfacade)**	roof surface (spatial polygons)	**10,071,755 roofs** 3,844,000 buildings (SB_UUID)	GeoPackage (gpkg) https://opendata.swiss/de/dataset/eignung-von-hausdachern-fur-die-nutzung-von-sonnenenergie	**Download: 1.5 GB gpkg.zip** parquet only: 0.77GB
solkat month	roof surface × month (table, no geometry)	**120,861,060 rows** (= 10,071,755 × 12)	GeoPackage (gpkg) https://opendata.swiss/de/dataset/eignung-von-hausdachern-fur-die-nutzung-von-sonnenenergie	**Download: 5.7 GB gpkg.zip** parquet only: 4.7 GB
gwr	building (EGID); Dwelling (apartments, EWID) (table, opt. spatial points)	**3,253,788 buildings** 5,199,292 dwellings	not on opendata.swiss (only cantonal copies); BFS: housing-stat.ch/de/madd/public.html	**Download: 0.95 GB ch.zip** parquet + GeoJSON: 4.8 GB
pv	production plant (spatial points)	**239,214 pv plants** (snapshot to Aug 2024)	GeoPackage (gpkg) https://opendata.swiss/de/dataset/elektrizitatsproduktionsanlagen	parquet + GeoJSON: 48 MB

How the four sources feed topo



02

Split And Partition Data Early!

Splitting solkat and processing it in tranches

1 Split geometry from attributes

Split data and geometry for all geo data frames for faster importing

DF_UID	SB_UUID	GWR_EGID	FLAECHE	...	geometry
d1		e1			⬠
d2		e1			⬠
d3		e2			⬠
d4		e3			⬠
d5		e3			⬠

all columns except geometry

DF_UID + geometry

DF_UID	SB_UUID	GWR_EGID	FLAECHE	...
d1		e1		
d2		e1		
d3		e2		
d4		e3		
d5		e3		

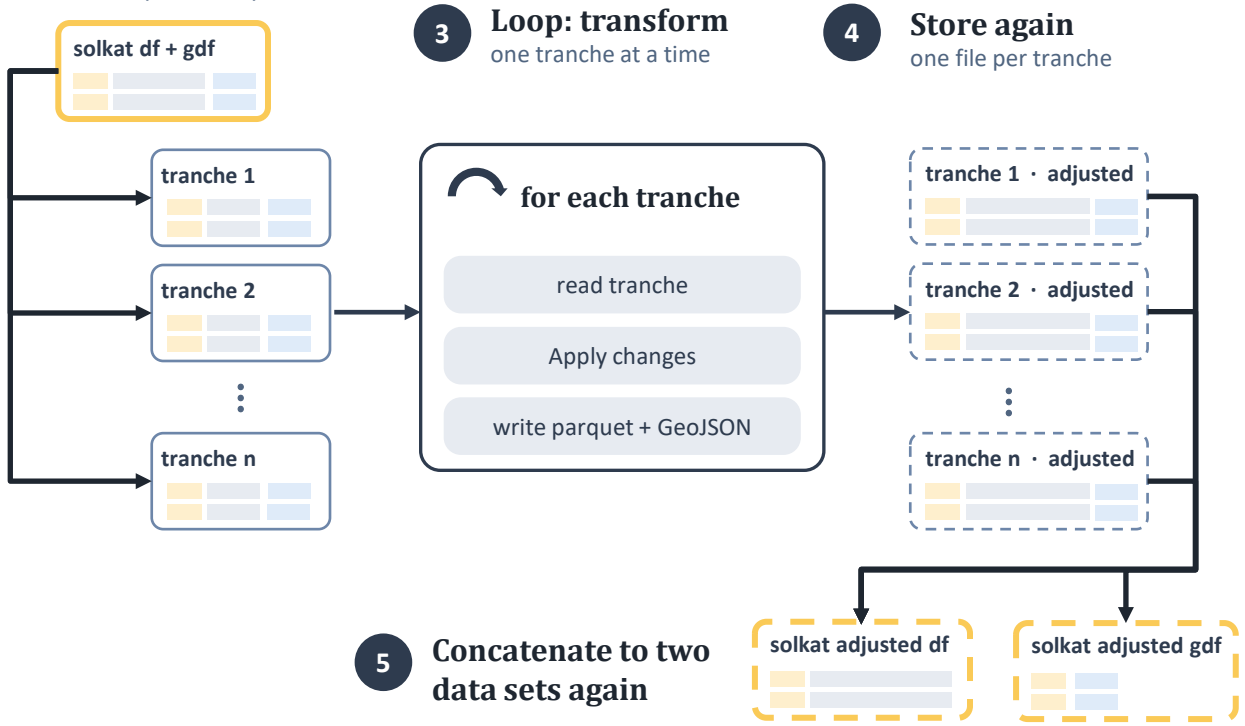
DF_UID	geometry
d1	⬠
d2	⬠
d3	⬠
d4	⬠
d5	⬠

solkat_pq.parquet
all attributes
parquet (polars / pandas)

solkat_geo.geojson
DF_UID, BFS_NUMMER, geometry
GeoJSON (GeoPandas)

2 Partition by EGID

100 EGIDs per tranche,
only necessary data



One static topo file, hourly profiles in tranches

1 Store gwr + solkat once

static data, one row per EGID - df_uid pair

gwr				solkat			
EGID	GKLAS	GAREA	...	df_uid	FLAECHE	AUSRICHTUNG	...
e1				d1			
e1				d2			
e2				d3			
e3				d4			
e3				d5			

× 8,760 hours

2 Hourly pv profiles in tranches

8,760 rows per EGID – df_uid pair

radiation
8,760 hours

solkat month
A, B, C per roof

tranche 1 · EGIDs 1–250

EGID	df_uid	t	pvprod_kW
e1	d1	t_1	
e1	d1	t_2	
⋮	⋮	⋮	⋮
e1	d1	t_8760	
e1	d2	t_1	

tranche 2
tranche 3
tranche 4
tranche 5
tranche n

3 Select EGIDs

grey: pv installed or grid node closed

gwr			solkat		
EGID			df_uid		
e1			d1		
e1			d2		
e2			d3		
e3			d4		
e4			d5		

4 Loop over tranches

one file at a time

for each tranche

tranche 1
tranche 2
tranche 3
tranche 4
tranche 5
tranche n

keep selected EGIDs

join static data

sum peak-hour pvprod

5 Collect results

same rows, extended by peak-hour production (kWh)

gwr			solkat			new
EGID			df_uid			peak pvprod
e1			d1			312
e1			d2			287
e2			d3			–
e3			d4			195
e4			d5			–

Why: static data is stored once in topo; hourly files keep only keys and hourly values and get the static data joined back when read.

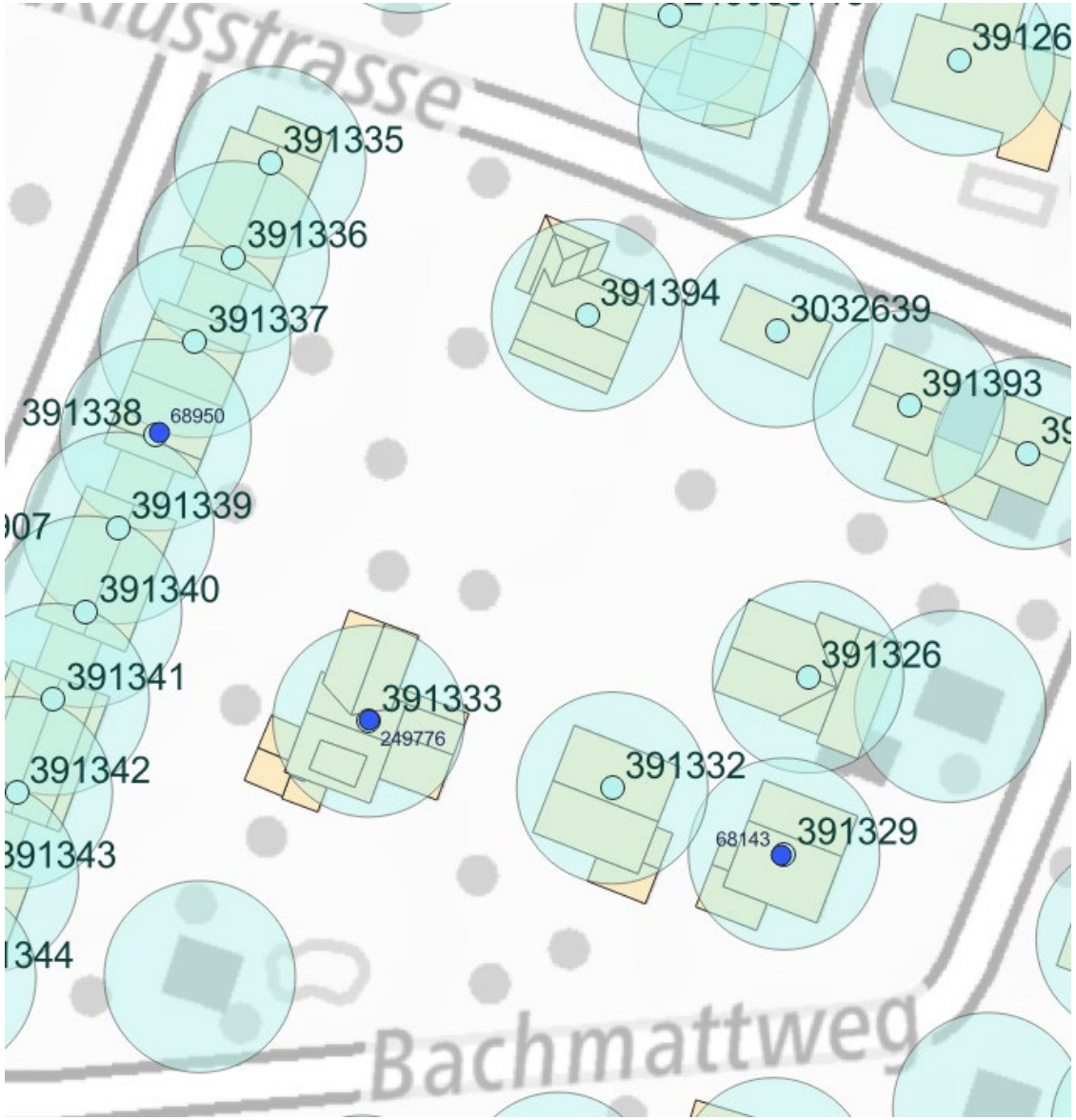
03

Current PV Installations



Spatial Join On Buffered EGIDs

- Spatial coordinates of PV installations quite good
- Buffered gwrEGID by 10 metters and assigned installtions
  - Single match -> intended case
  - Multiple matches -> to closest gwrEGID within match



03

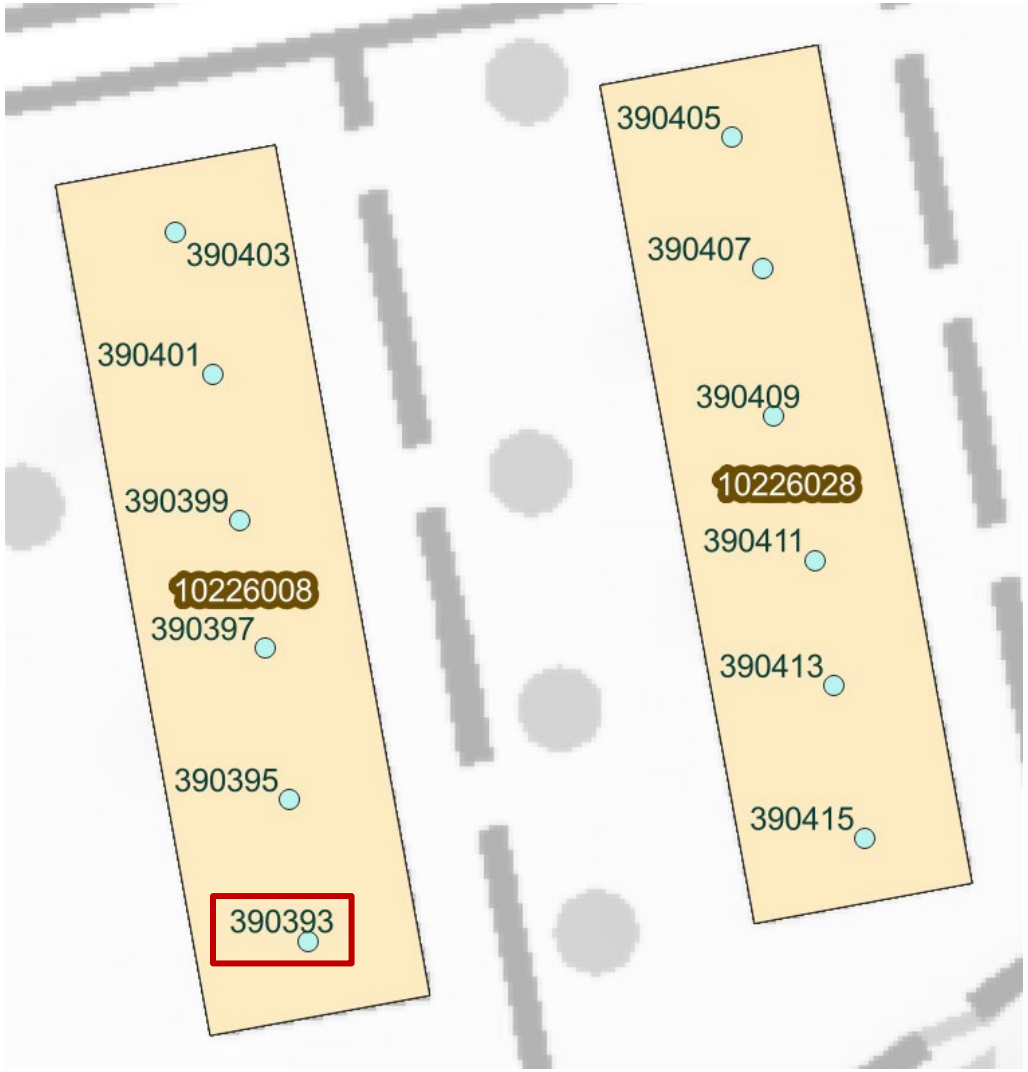
«EGID» Mismatch

gwr[EGID] =/= solkat[GWR_EGID]

Single solkat Polygon, But Multiple EGIDs

egid_join_union
shape: (6, 4)

EGID_old_solkat	DF_UID	index_right	EGID_gwradded
str	str	f64	str
390393	10226008	664.0	390393
390393	10226008	2123.0	390395
390393	10226008	1455.0	390397
390393	10226008	1014.0	390399
390393	10226008	257.0	390401
390393	10226008	1812.0	390403



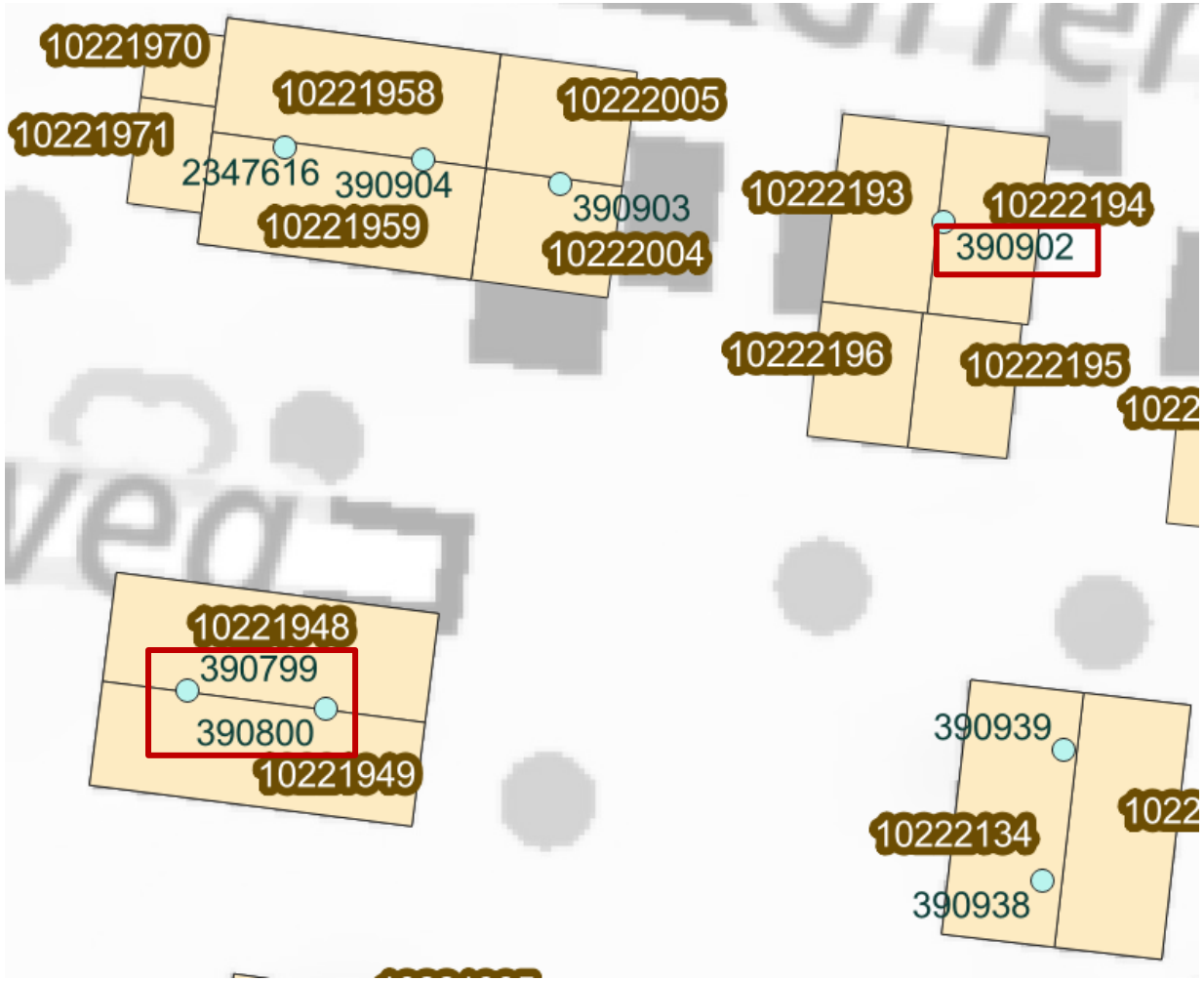
Multiple solkat Polygons

egid_join_union
shape: (2, 4)

EGID_old_solkat	DF_UID	index_right	EGID_gwradded
str	str	f64	str
390799	10221948_10221949	88.0	390800
390799	10221948_10221949	1932.0	390799

egid_join_union
shape: (1, 4)

EGID_old_solkat	DF_UID	index_right	EGID_gwradded
str	str	f64	str
390902	10222193_10222194_10222195_102…	907.0	390902

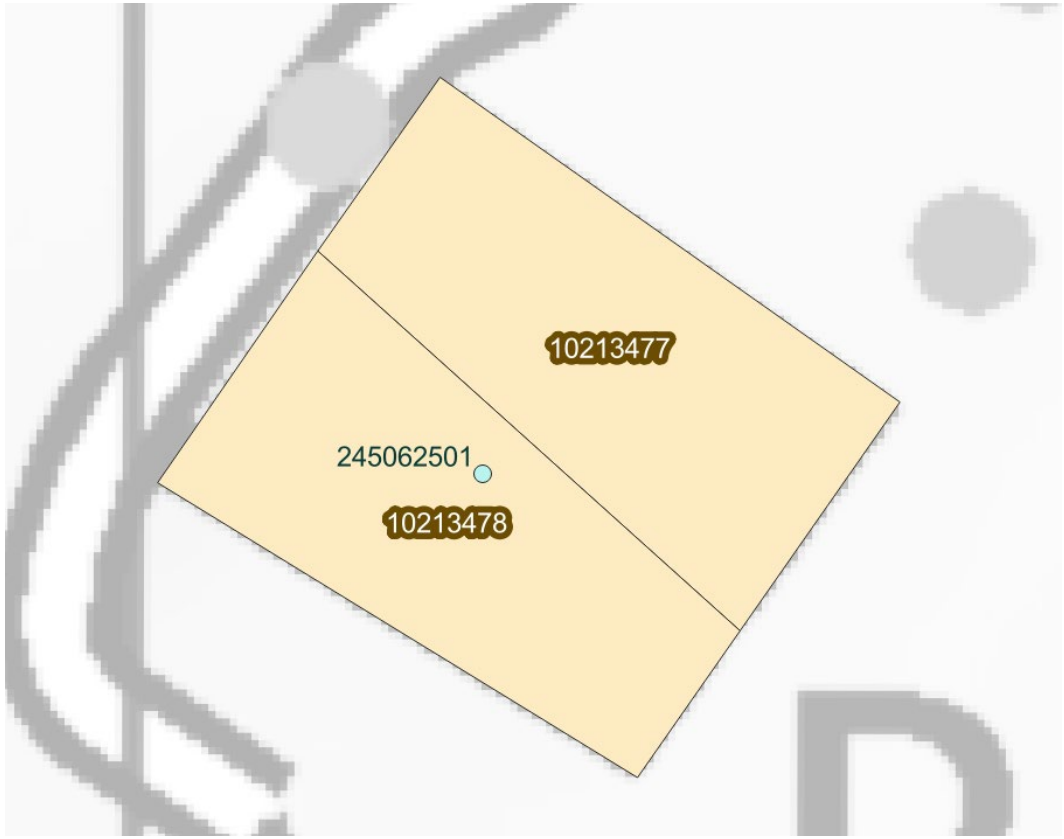


Missing solkat.GWR_EGID

egid_join_union
shape: (1, 4)

EGID_old_solkat	DF_UID	index_right	EGID_gwradded
str	str	f64	str
egid_proxysbuuid_{2ADC55D1-CB4…	10213477_10213478	276.0	245062501

Taking SB_UUID instead of GWR_EGID as a union proxy

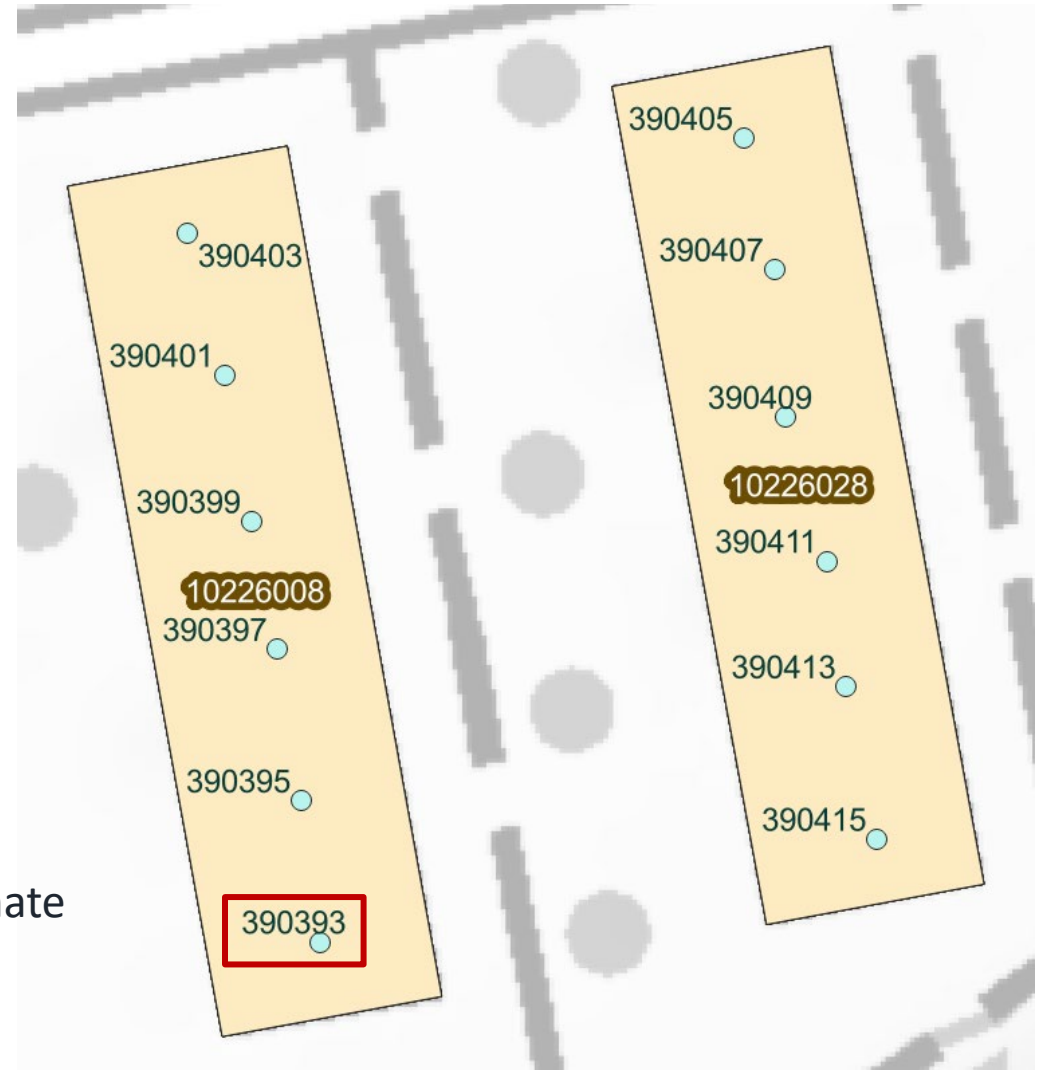


solkat Adjustments

- Copy and concatenate solkat partitiones for n EGIDs
- Set gwr based EGID as actual EGIDs
- Devide relevant columns (FLAECHE, STROMERTRAG etc.) by n EGIDs in shape

gwr				solkat			
EGID	GKLAS	GAREA	...	df_uid	FLAECHE	AUSRICHTUNG	...
e1				d1			
e2				d1			
e3				d1			
e4				d1			
e5				d1			

Copy & concatenate



Watch Out: solkat

- Take care when matching solkat based time series data to the adjusted static solkat df
  - can lead to unintended duplication (nEGIDs x overestimation of PV production.

- DF_UID indentifier needs to remain «untouched» for solkat_month matching

gwr				solkat			
EGID	GKLAS	GAREA	...	df_uid	FLAECHE	AUSRICHTUNG	...
e1				d1			
e2				d1			
e3				d1			
e4				d1			
e5				d1			

03

GWR data

gwr

Watch Out: gwr

- Inconsistencies accross features (e.g. GBAUJ)
  - Silently drops certain EGIDs
  - Workarounds (building status GSTAT more accurate than GBAUJ)
- Two data sources
  - «Single» Buildings (EGID)
  - Multi-Apartment / Dwellings in Buildings (EGID + EWID)
    - EWID registery features EGID, so within apartment destinction possible
    - Also here inconsistencies accross features

=> Use standardized checksums to ensure you don’t miss too many buildings along the way

